# Technical Supplement

## SABER: Support-Conditioned Adaptive Bellman Error Reweighting for Sparse-Reward Hindsight Experience Replay

Anonymous Submission

**Overview.** This supplement gives a complete mathematical specification and pseudocode for SABER, followed by experimental settings, seed-level primary results, sensitivity and scope controls, support-signal diagnostics, runtime measurements, and an index of the anonymous code-and-data artifacts.

**Working-draft completion status.** One result slot remains *pending*: FPP DDPG Actor+Critic. No value is inferred from its incomplete run. This marker and paragraph must be removed after a complete, validated five-seed artifact replaces the remaining slot.

## 1 Complete SABER Specification

### 1.1 Scope, Problem Setting, and Notation

SABER is a critic-loss allocation rule for replay-based off-policy actor–critic learning. It is neither a replay sampler nor a network module. Consider a finite-horizon goal-conditioned MDP

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{G}, P, r, \gamma, H).$$

The agent receives observation $o_t$, desired goal $g$, and network input $x_t = (o_t, g)$. The achieved-goal map is $m : \mathcal{S} \to \mathcal{G}$. The sparse reward used in the experiments is

$$r(s_t, a_t, s_{t+1}, g) = \begin{cases} 0, & d(m(s_{t+1}), g) \le \epsilon_g, \\ -1, & \text{otherwise.} \end{cases} \tag{1}$$

Replay stores the executed action $a_i^r$. We reserve $a_i^\pi$ for the detached current-policy action used only by SABER's discrete support query. There are $n_Q = 1$ critics for DDPG and $n_Q = 2$ for TD3 and SAC.

### 1.2 Future-Goal HER Batch and Unchanged Bellman Targets

Let an episode contain states $s_0, \dots, s_T$ and transitions indexed by $t = 0, \dots, T-1$. When transition $t$ is selected for relabeling and has an eligible later stored transition, the implemented future-goal convention samples

$$\begin{aligned} u &\sim \mathrm{Unif}\{t+1, \dots, T-1\}, \\ \widetilde{g}_i &= m(s_{u+1}), \\ \widetilde{r}_i &= r(s_t, a_t, s_{t+1}, \widetilde{g}_i). \end{aligned} \tag{2}$$

If the eligible set is empty, or if relabeling is not selected, the original desired goal and its corresponding reward are retained. Thus the nominal relabeling probability is $p_{\mathrm{HER}} = 0.8$, while the realized fraction can be slightly lower near an episode boundary. The relabeled network inputs are

$$x_i = (o_i, \widetilde{g}_i), \qquad x_i' = (o_i', \widetilde{g}_i).$$

Let $\chi_i \in \{0, 1\}$ be the bootstrap mask. Successful relabeled transitions and time-limit transitions use $\chi_i = 1$; genuine terminating transitions use $\chi_i = 0$. For backbone $\mathsf{A}$, define

$$y_i^{\mathsf{A}} = \widetilde{r}_i + \gamma \chi_i V_i^{\mathsf{A}}(x_i'), \tag{3}$$

where target values are evaluated without gradient. Writing barred parameters for target networks, the implemented continuation values have the standard forms

$$V_i^{\mathrm{DDPG}} = Q_{\bar{\phi}_1}(x_i', \mu_{\bar{\theta}}(x_i')), \tag{4}$$

$$\begin{aligned} \varepsilon_i &\sim \mathrm{clip}\big(\mathcal{N}(0, \sigma_{\mathrm{targ}}^2 I), -c, c\big), \\ \widetilde{a}_i' &= \mathrm{clip}_{\mathcal{A}}(\mu_{\bar{\theta}}(x_i') + \varepsilon_i), \\ V_i^{\mathrm{TD3}} &= \min_{k \in \{1,2\}} Q_{\bar{\phi}_k}(x_i', \widetilde{a}_i'), \end{aligned} \tag{5}$$

$$\begin{aligned} a_i' &\sim \pi_\theta(\cdot \mid x_i'), \\ V_i^{\mathrm{SAC}} &= \min_{k \in \{1,2\}} Q_{\bar{\phi}_k}(x_i', a_i') - \alpha \log \pi_\theta(a_i' \mid x_i'). \end{aligned} \tag{6}$$

SABER does not change any quantity in Eqs. (2)–(6). For critic $k$, the replay-action Bellman residual remains

$$\delta_{ik} = Q_{\phi_k}(x_i, a_i^r) - y_i^{\mathsf{A}}. \tag{7}$$

Under ordinary uniformly sampled HER, every squared residual has explicit coefficient one.

### 1.3 Current-Policy Local Support

For each relabeled sample, SABER forms the full discrete observation–goal–horizon context

$$c_i = C(o_i, \widetilde{g}_i, h_i), \tag{8}$$

where $h_i$ is the remaining horizon. A lower-dimensional retrieval key $C^{\mathrm{ret}}(o_i, h_i)$ is used only to query the replay index. Let $\mathbf{J}_i^{\mathrm{ret}}$ be the resulting ordered list of currently valid, unique replay slots. Invalid entries and duplicate slot identifiers are removed in maintained index order, retaining the first valid occurrence. The bounded scan is

$$S_i = \mathrm{Tail}_{N_{\mathrm{scan}}}\left(\mathbf{J}_i^{\mathrm{ret}}\right), \qquad |S_i| \le N_{\mathrm{scan}}. \tag{9}$$

The tail refers to deterministic maintained index order, not guaranteed temporal recency after circular-buffer wraparound. The coarse key is only an acceleration device. Support eligibility is determined by exact component-wise equality of the full context:

$$I_i = \{j \in S_i : C(o_j, \widetilde{g}_i, h_j) = C(o_i, \widetilde{g}_i, h_i)\}. \tag{10}$$

The query's relabeled goal $\widetilde{g}_i$ is used on both sides; a candidate's originally stored desired goal is not used for matching. The sampled transition is not removed from consideration and may therefore occur in $I_i$.

For action dimension $d$, let $\ell_d$ and $u_d$ be its environment bounds. SABER first maps the action to

$$z_d(a) = \mathrm{clip}\left(\frac{a_d - \ell_d}{u_d - \ell_d}, 0, 1\right) \tag{11}$$

and assigns the discrete component

$$b_{K,d}(a) = \min\{K - 1, \lfloor K z_d(a) \rfloor\}. \tag{12}$$

The multidimensional bin $b_K(a)$ is the tuple of these components; the implementation stores its equivalent base-$K$ integer encoding. The local set of occupied replay-action bins is

$$\mathcal{U}_i = \left\{b_K(a_j^r) : j \in I_i\right\}, \qquad n_i = |I_i|. \tag{13}$$

The binary support indicator is

$$\psi_i = \mathbf{1}\{n_i \ge N_{\mathrm{local}}\}\, \mathbf{1}\{b_K(a_i^\pi) \in \mathcal{U}_i\}. \tag{14}$$

If retrieval returns no candidates, exact matching yields too few candidates, or the policy bin is unoccupied, then $\psi_i = 0$. Main SABER uses

$$K = 5, \qquad N_{\mathrm{scan}} = 512, \qquad N_{\mathrm{local}} = 4.$$

DDPG and TD3 use their deterministic actor output as $a_i^\pi$. SAC uses the deterministic squashed actor mean for this query only; its target and actor updates continue to use stochastic policy samples. No gradient passes through $a_i^\pi$, action binning, retrieval, or $\psi_i$.

| Task family | Full context components | Coarse retrieval key |
|---|---|---|
| FetchPickAndPlace | Phase, hand–object distance, object–goal distance, object height, gripper gap, horizon | Hand–object distance, object height, gripper gap, horizon |
| DrawerOpen/Close | Phase, hand–handle distance, handle–goal distance, handle $y$, gripper state, horizon | Hand–handle distance, handle $y$, gripper state, horizon |

**Table 1:** Task-specific support contexts. Candidate retrieval uses the coarse key, after which exact component-wise equality on the full discretized context defines the local replay set.

| Component | FetchPickAndPlace thresholds | Drawer thresholds |
|---|---|---|
| Phase | approach, grasp, lift, place | approach, contact, opening/closing, open/closed |
| Hand distance | .03, .05, .08, .12, .20, .40 | .03, .05, .08, .12, .20, .40 |
| Goal distance | .05, .10, .20, .40, .70 | .03, .05, .08, .12, .20, .40, .70 |
| Task coordinate | object $z$: .43, .48, .55, .65, .75 | handle $y$: $-.05, 0, .05, .10, .15, .25$ |
| Gripper | gap: .02, .05, .08 | state: $-.50, 0, .50$ |
| Remaining horizon | $10, 25, 50, 100$ | $10, 25, 50, 100, 200, 500$ |

**Table 2:** Discretization boundaries used to construct support contexts. Values are applied component-wise in environment units. The main method normalizes each action dimension to its action bounds and uses $K = 5$ bins per dimension, $N_{\mathrm{scan}} = 512$, $N_{\mathrm{local}} = 4$, and $\lambda_S = 1$.

For the Drawer phase definitions, let $d_h$ be hand–handle distance and $d_g$ be handle–goal distance. The terminal phase is open or closed when $d_g \le 0.05$; opening or closing requires $d_h \le 0.08$ and $0.05 < d_g \le 0.12$; contact requires $d_h \le 0.08$ and $d_g > 0.12$; remaining samples are approach. Phases use the relabeled training goal.

### 1.4 Mean-Normalized Critic-Only Objective

SABER assigns the raw coefficient

$$w_i^{\mathrm{raw}} = 1 + \lambda_S \psi_i \tag{15}$$

and normalizes it by the minibatch mean:

$$\bar{w}_B = \frac{1}{|B|} \sum_{j \in B} w_j^{\mathrm{raw}}, \qquad \Omega_i = \frac{w_i^{\mathrm{raw}}}{\max(\epsilon_w, \bar{w}_B)}. \tag{16}$$

We use $\lambda_S = 1$ and $\epsilon_w = 10^{-8}$. The safeguard is inactive because $w_i^{\mathrm{raw}} \geq 1$. If $p_B = |B|^{-1} \sum_i \psi_i$, then

$$\Omega_i = \begin{cases} \dfrac{1 + \lambda_S}{1 + \lambda_S p_B}, & \psi_i = 1, \\ \dfrac{1}{1 + \lambda_S p_B}, & \psi_i = 0. \end{cases} \tag{17}$$

Hence every sample remains in the update with a positive coefficient, the minibatch mean coefficient is one, and support-positive samples receive relative weight $1 + \lambda_S$ compared with support-negative samples. If all indicators in a minibatch are equal, then every normalized coefficient is exactly one.

The critic objective is

$$\mathcal{L}_Q^{\mathrm{SABER}} = \frac{1}{|B|} \sum_{i \in B} \Omega_i \sum_{k=1}^{n_Q} \delta_{ik}^2. \tag{18}$$

For TD3 and SAC, the same $\Omega_i$ multiplies both critic residuals. The main variant is critic-only:

$$\mathcal{L}_\pi^{\mathrm{SABER}} = \mathcal{L}_\pi^{\mathsf{A}}. \tag{19}$$

Mean-one normalization preserves the average explicit coefficient, but does not imply an unchanged numerical loss or gradient norm because support may correlate with residual magnitude.

### 1.5 End-to-End Update

Algorithm 1 states one complete learning update. The replay index is maintained as transitions enter or overwrite replay slots. Candidate retrieval supplies only a bounded list; exact full-context matching determines support eligibility.

**Algorithm 1: One SABER update**

**Require:** replay $\mathcal{D}$ and index $\mathcal{R}$; actor $\pi_\theta$; critics $\{Q_{\phi_k}\}_{k=1}^{n_Q}$; target networks
**Require:** context maps $C, C^{\mathrm{ret}}$; $K, N_{\mathrm{scan}}, N_{\mathrm{local}}, \lambda_S$
$B \leftarrow \mathrm{UniformSample}(\mathcal{D})$
Apply Eq. (2) where selected and eligible; recompute rewards.
**for each** relabeled sample $i \in B$ **do**
  Construct $x_i, x_i'$, bootstrap mask $\chi_i$, and detached $y_i^{\mathsf{A}}$ using Eqs. (3)–(6).
  $a_i^\pi \leftarrow \mathrm{sg}[\mathrm{QueryAction}_{\mathsf{A}}(x_i)]$
  $q_i^{\mathrm{ret}} \leftarrow C^{\mathrm{ret}}(o_i, h_i)$
  $\mathbf{J}_i^{\mathrm{ret}} \leftarrow \mathcal{R}[q_i^{\mathrm{ret}}]$
  Remove invalid and duplicate slots from $\mathbf{J}_i^{\mathrm{ret}}$.
  $S_i \leftarrow \mathrm{Tail}_{N_{\mathrm{scan}}}(\mathbf{J}_i^{\mathrm{ret}})$
  $I_i \leftarrow \{j \in S_i : C(o_j, \widetilde{g}_i, h_j) = C(o_i, \widetilde{g}_i, h_i)\}$
  $\mathcal{U}_i \leftarrow \{b_K(a_j^r) : j \in I_i\}$
  $\psi_i \leftarrow \mathbf{1}\{|I_i| \geq N_{\mathrm{local}}\}\mathbf{1}\{b_K(a_i^\pi) \in \mathcal{U}_i\}$
  $w_i^{\mathrm{raw}} \leftarrow 1 + \lambda_S \psi_i$
**end for**
$\Omega_i \leftarrow w_i^{\mathrm{raw}} / \max(\epsilon_w, |B|^{-1} \sum_{j \in B} w_j^{\mathrm{raw}})$ for every $i$.
Compute replay-action residuals using Eq. (7).
Update all critics with Eq. (18).
Apply the backbone's unchanged actor, temperature, delayed-update, and target-network rules.

The algorithm introduces no auxiliary target and never regresses the critic at $a_i^\pi$. It does not alter minibatch membership, HER relabeling, reward recomputation, exploration, network architecture, optimizer, or UTD. The indicator is a binary occupancy proxy, not an estimate of density, uncertainty, target accuracy, target variance, or TD-error rank. Finally, SABER cannot mark an action as locally supported unless its action bin is already represented by replay actions in the matched context.

## 2 Experimental and Reproducibility Details

### 2.1 Tasks, Cohorts, and Shared Training Settings

**Purpose.** This section fixes the experimental scope before any method is compared. Its purpose is to identify which quantities are paired, which quantities are shared, and which task–backbone combinations support a matched SABER claim.

**Protocol.** Tables 3–5 give the settings shared within each matched cohort. The primary comparison contains seven task–backbone cohorts and uses paired seeds 1–5. Drawer SAC is not a primary cohort because a complete matched HER–SABER pair was not collected.

| Task | Horizon | Budget | Backbones | Exact primary grid |
|---|---|---|---|---|
| FetchPickAndPlace-v4 | 50 | 1M | DDPG, TD3, SAC | 10k (DDPG/TD3), 50k (SAC) |
| DrawerOpen-v3 | 200 | 2M | DDPG, TD3 | 10k |
| DrawerClose-v3 | 200 | 2M | DDPG, TD3 | 10k |

**Table 3:** Primary task and cohort definitions. Every method within a task–backbone cohort uses the same environment-step budget and exact evaluation grid. Primary comparisons use paired seeds 1–5.

| Setting | Value or convention |
|---|---|
| HER | Future-goal convention in Sec. 1.2; nominal $p_{\mathrm{HER}} = 0.8$ |
| Replay | Uniform for HER and SABER; capacity 300k; minibatch size 256 |
| Networks | Actor and critic MLPs with two hidden layers of 256 units |
| Optimization | Adam, actor and critic learning rates $3 \times 10^{-4}$ |
| Discount and targets | $\gamma = 0.98$, Polyak coefficient $\tau = 0.005$ |
| Update schedule | UTD 1; learning begins after 5k steps; random actions for first 10k steps |
| Termination | Successful relabeled transitions are nonterminal; time limits bootstrap |
| Evaluation | Deterministic policy, 20 episodes per checkpoint |
| Statistics | Seed mean and sample SD; paired percentile-bootstrap CI, 10,000 resamples, RNG seed 2027 |
| Curve processing | Exact common checkpoints only; no smoothing, interpolation, extrapolation, or filling |

**Table 4:** Shared training, evaluation, and statistical settings. Matched HER–SABER runs use identical replay and relabeling code.

| Backbone | Critic | Exploration or entropy | Backbone-specific settings |
|---|---|---|---|
| DDPG | One critic | Gaussian action noise, SD $0.2$ | Deterministic actor |
| TD3 | Twin critics | Gaussian action noise, SD $0.2$ | Target noise $0.2$, clip $0.5$, policy delay 2 |
| SAC | Twin critics | Automatic entropy tuning | Initial $\alpha = 0.2$, $\alpha$ LR $3 \times 10^{-4}$, target entropy $-4$ |

**Table 5:** Backbone-specific settings. SABER preserves each backbone's target construction and actor-update rule.

**Interpretation boundary.** Within each paired cohort, HER and SABER share the task, seed, training budget, replay and relabeling code, backbone, optimizer, exploration process, UTD, and evaluation grid. Their intended algorithmic difference is therefore the per-sample critic-loss coefficient. Absolute AUC values are not compared across tasks because horizons and training budgets differ.

### 2.2 Comparison Methods

**Purpose.** The comparison suite separates three operational intervention points after experience collection: hindsight selection, replay-transition selection, and fixed-minibatch critic-loss allocation.

| Method | Selection changed | Loss coefficient | Correction |
|---|---|---|---|
| HER | None beyond uniform replay and HER relabeling | Uniform | None |
| SABER | None | Current-policy local-support weight | Unit minibatch mean |
| HGR | Episode and transition–future-goal pair | HGR importance weight | Two-stage IS |
| PER | Replay transition | PER importance weight | Annealed IS |
| TD-error weighting | None | TD-error-derived weight | Unit minibatch mean |

**Table 6:** Operational intervention points. HGR uses priority exponents $\alpha_{\mathrm{episode}} = \alpha_{\mathrm{pair}} = 0.6$, while PER uses $\alpha = 0.6$. Both anneal $\beta$ linearly from $0.4$ to $1.0$ over critic-update progress and use priority offset $10^{-6}$. TD-error weighting retains uniform replay and is therefore the closest allocation-stage control.

**Protocol.**

**Prioritization controls.** HGR prioritizes an episode and then a transition–future-goal pair, multiplies the two normalized importance factors on relabeled samples, and leaves the original-goal coefficient at one. PER samples replay transitions with TD-error priority and annealed importance correction. Both change realized selection, unlike SABER and the TD-error critic-loss weighting control.

**TD-error critic-loss weighting.** This control preserves uniform replay and changes only normalized per-sample critic coefficients. DDPG uses its single absolute TD error. SAC uses $\min(|\delta_{i1}|, |\delta_{i2}|)$ as the signal and applies the resulting coefficient to both critic losses. TD3 results are not reported because the stored control used independent target-policy smoothing noise for weight construction and optimized residuals, preventing the intended matched-signal interpretation.

**Interpretation boundary.** HER is the direct matched baseline. HGR and PER are broader prioritization comparisons because they alter realized selection, whereas TD-error weighting is the closest alternative allocation-signal control where validated. Thus an ordering against HGR or PER does not isolate loss allocation alone, while an ordering against TD-error weighting does not establish superiority over every possible weighting signal.

### 2.3 Metrics and Statistical Analysis

**Purpose.** The metrics distinguish full-training performance, late-training performance, final-checkpoint behavior, and threshold-reaching time without smoothing away seed variability.

**Protocol.** Let $q_j \in [0, 1]$ be deterministic evaluation success at exact shared checkpoint $t_j$. Normalized success AUC is

$$\mathrm{AUC} = \frac{1}{t_m - t_1} \sum_{j=1}^{m-1} \frac{q_j + q_{j+1}}{2} (t_{j+1} - t_j).$$

Final-quarter AUC applies the same calculation to 750k–1M for FPP and 1.5M–2M for Drawer tasks. Final success is the value at the final checkpoint. Hit90 is the first exact checkpoint with $q_j \geq 0.9$; its median is computed only over seeds that reach the threshold, so reached-seed coverage is always reported.

For paired comparisons, each seed contributes one method difference. The reported 95% intervals are percentile intervals from 10,000 paired bootstrap resamples with pseudorandom seed 2027. Sample standard deviations use $n - 1$ in the denominator. No curve is smoothed, interpolated, extrapolated, or forward-filled.

**Interpretation boundary.** Where reported, Steps/s is the arithmetic mean over seeds 1–5 of each run's completed environment steps divided by total training runtime. These historical measurements were not hardware-normalized across run families and are therefore descriptive implementation measurements, not learning-efficiency claims. Control and reference tables use the common order Cohort, Method/Variant, AUC, FQ-AUC, Final, Hit90, and Steps/s; diagnostic tables retain signal-specific columns.

## 3 Primary Comparison Details

### 3.1 Drawer Learning Curves

**Purpose.** Full-span AUC compresses the learning trajectory into one number. The Drawer curves expose when success emerges, whether gains persist late in training, and how much of each aggregate ordering is driven by seed variability.

**Results.** DrawerOpen DDPG shows the clearest separation from HER, whereas the TD3 cohorts and comparisons with HGR and PER exhibit larger seed dispersion and mixed metric-dependent orderings. On DrawerClose DDPG, HGR leads full-span AUC but not FQ-AUC, illustrating that early and late summaries can disagree.

**Interpretation.** The trajectories support the matched HER result but do not establish a universal ranking against HGR or PER. They also show why full-span AUC, FQ-AUC, and final success are retained separately: early learning, late occupancy, and a single final checkpoint need not order methods identically.

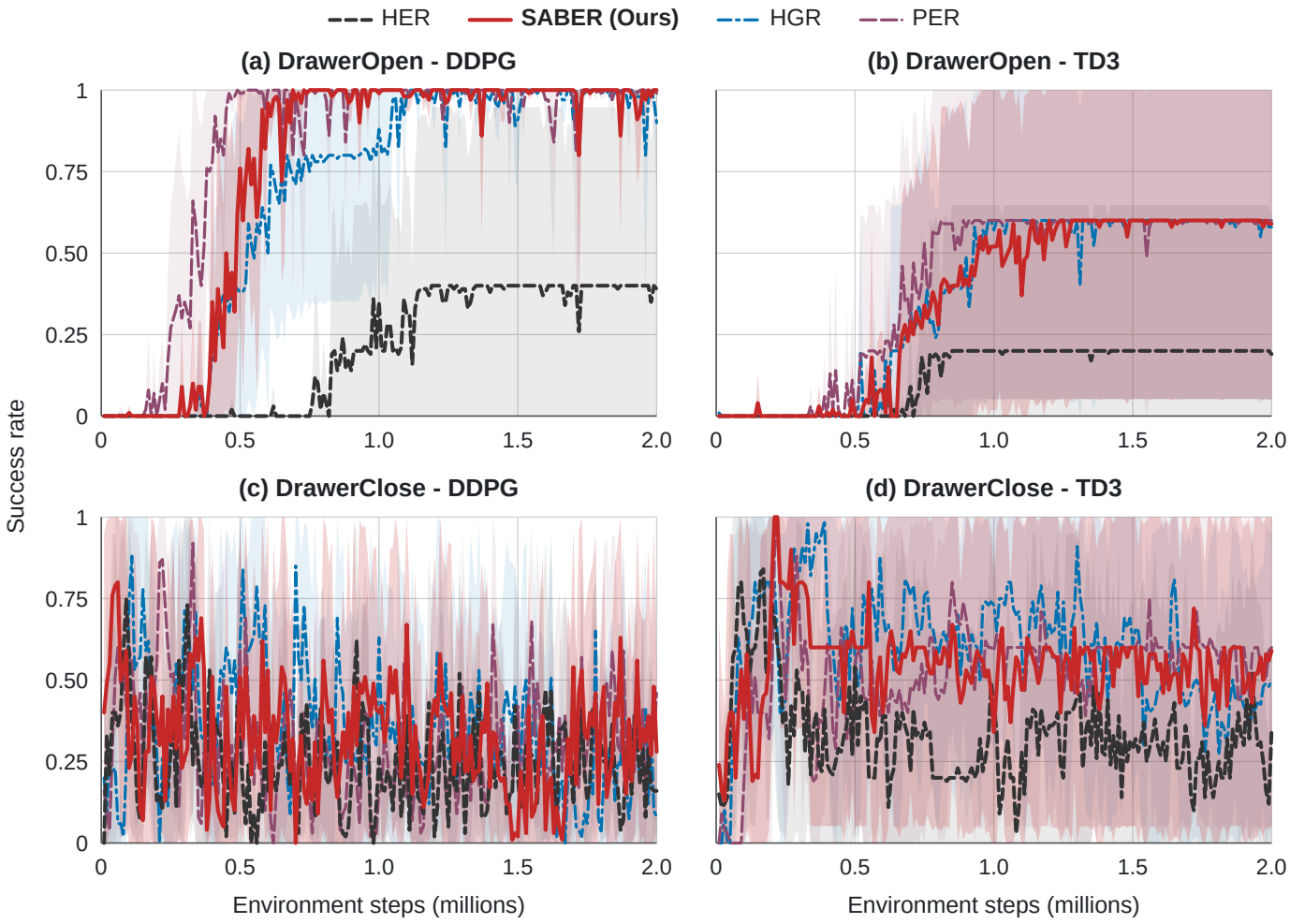



**Figure 1:** DrawerOpen and DrawerClose learning curves for DDPG and TD3. Lines are seed means and bands are sample standard deviations over paired seeds 1–5 on the exact 10k grid. No smoothing or interpolation is applied.

### 3.2 Time to Threshold

**Purpose.** Hit90 tests whether a method reaches a practically high success level and how early it first does so, complementing metrics that average over checkpoints.

**Results.** Across the three FPP backbones, SABER reaches Hit90 in all 15 runs, compared with 11/15 for HER, 5/15 for HGR, and 10/15 for PER. On DrawerOpen DDPG, SABER reaches in 5/5 seeds at a reached-seed median of 500k, whereas HER reaches in 2/5 at 960k. On DrawerClose, every primary method reaches Hit90 in all seeds for both backbones, despite substantial differences in later performance.

**Interpretation.** Hit90 is a threshold-crossing summary rather than a stability measure. Its conditional median is meaningful only together with reached-seed coverage, and early threshold crossing cannot substitute for FQ-AUC or final success.

| Task | Backbone | Method | Median Hit90 | Reached seeds | Steps/s |
|---|---|---|---|---|---|
| FetchPickAndPlace | DDPG | SABER | 540k | 5/5 | 41.7 |
| | | HER | 600k | 5/5 | 65.0 |
| | | HGR | 930k | 2/5 | 16.9 |
| | | PER | 640k | 4/5 | 58.0 |
| FetchPickAndPlace | TD3 | SABER | 840k | 5/5 | 41.2 |
| | | HER | 980k | 3/5 | 65.3 |
| | | HGR | 630k | 1/5 | 12.9 |
| | | PER | 1.00M | 1/5 | 50.8 |
| FetchPickAndPlace | SAC | SABER | 750k | 5/5 | 29.1 |
| | | HER | 800k | 3/5 | 38.9 |
| | | HGR | 800k | 2/5 | 8.4 |
| | | PER | 550k | 5/5 | 34.8 |
| DrawerOpen | DDPG | SABER | 500k | 5/5 | 23.3 |
| | | HER | 960k | 2/5 | 41.1 |
| | | HGR | 530k | 5/5 | 13.2 |
| | | PER | 330k | 5/5 | 34.0 |
| DrawerOpen | TD3 | SABER | 780k | 3/5 | 21.6 |
| | | HER | 740k | 1/5 | 41.5 |
| | | HGR | 760k | 3/5 | 11.6 |
| | | PER | 720k | 3/5 | 34.6 |
| DrawerClose | DDPG | SABER | 20k | 5/5 | 24.0 |
| | | HER | 70k | 5/5 | 41.2 |
| | | HGR | 100k | 5/5 | 13.2 |
| | | PER | 60k | 5/5 | 33.9 |
| DrawerClose | TD3 | SABER | 80k | 5/5 | 22.5 |
| | | HER | 70k | 5/5 | 41.7 |
| | | HGR | 80k | 5/5 | 11.8 |
| | | PER | 120k | 5/5 | 34.4 |

**Table 7:** Hit90 for the primary comparisons. Hit90 is the first exact-grid checkpoint at which deterministic evaluation success reaches at least 0.9. Medians use only reached seeds, so reached-seed coverage is reported separately. Steps/s is the descriptive seed mean defined in Section 1.3.

### 3.3 FPP TD3 Confirmatory and HER-Indexing Sensitivity Results

**Purpose.** The ten-seed extension tests whether the primary five-seed FPP TD3 direction persists under a larger frozen cohort. The indexing sensitivity asks whether that direction is specific to excluding $m(s_{t+1})$ from future goals.

**Results.** In the ten-seed cohort, SABER exceeds HER in 8/10 pairs, with mean AUC difference $+0.0467$, paired-bootstrap 95% CI $[0.0085, 0.0783]$, and exact sign-flip $p = 0.0527$. Under inclusive-future indexing, the five-seed mean difference is $+0.0674$, with 3/5 positive pairs, 95% CI $[-0.0068, 0.1416]$, and sign-flip $p = 0.25$, compared with $+0.0708$ under the primary convention.

**Interpretation.** The extension narrows the estimated TD3 gain while preserving positive paired bootstrap interval evidence. The inclusive variant preserves the observed mean direction, but it is a five-seed sensitivity rather than an equivalence test between indexing conventions.

| Seed | HER AUC | SABER AUC | Difference |
|---|---|---|---|
| 1 | 0.3465 | 0.4301 | $+0.0836$ |
| 2 | 0.3727 | 0.4568 | $+0.0841$ |
| 3 | 0.3942 | 0.4960 | $+0.1018$ |
| 4 | 0.4109 | 0.4066 | $-0.0043$ |
| 5 | 0.3338 | 0.4225 | $+0.0886$ |
| 6 | 0.4301 | 0.5207 | $+0.0907$ |
| 7 | 0.3995 | 0.4614 | $+0.0619$ |
| 8 | 0.3068 | 0.3543 | $+0.0475$ |
| 9 | 0.4106 | 0.4114 | $+0.0008$ |
| 10 | 0.4210 | 0.3331 | $-0.0879$ |
| Mean | 0.3826 | 0.4293 | $+0.0467$ |

**Table 8:** Frozen FPP TD3 confirmatory extension over ten paired seeds. AUC is computed on the exact 10k–1M evaluation grid without smoothing, interpolation, or filling.

| Seed | HER AUC | SABER AUC | Difference |
|---|---|---|---|
| 1 | 0.3005 | 0.4576 | $+0.1571$ |
| 2 | 0.2674 | 0.2669 | $-0.0005$ |
| 3 | 0.3467 | 0.5179 | $+0.1712$ |
| 4 | 0.3487 | 0.4003 | $+0.0515$ |
| 5 | 0.3669 | 0.3247 | $-0.0422$ |
| Mean | 0.3261 | 0.3935 | $+0.0674$ |

**Table 9:** FPP TD3 sensitivity to inclusive future-goal indexing, which also admits $m(s_{t+1})$. AUC uses the exact 10k–1M evaluation grid. The final-quarter AUC difference is $+0.1854$ (95% CI $[-0.0212, 0.3920]$), and the final-success difference is $+0.2200$ (95% CI $[-0.2000, 0.5200]$).

# 4 Additional Controls and Sensitivities

## 4.1 Allocation Signal

**Purpose.** This control asks whether SABER benefits merely from assigning unequal critic coefficients, or whether current-policy local support provides a different allocation signal from TD error. Both methods retain uniform replay and change only normalized critic-loss coefficients.

| Cohort | Method/Variant | AUC | FQ-AUC | Final | Hit90 | Steps/s | $\Delta$AUC |
|---|---|---|---|---|---|---|---|
| FPP–DDPG | SABER (Main) | $0.609 \pm 0.038$ | $0.961 \pm 0.014$ | $0.960 \pm 0.042$ | 5/5 (0.540M) | 41.7 | $--$ |
| | TD-error weighting | $0.568 \pm 0.069$ | $0.910 \pm 0.119$ | $0.930 \pm 0.104$ | 5/5 (0.630M) | 61.1 | $+0.041$ $[-0.018, 0.123]$ |
| DrawerOpen–DDPG | SABER (Main) | $0.754 \pm 0.044$ | $0.987 \pm 0.018$ | $1.000 \pm 0.000$ | 5/5 (0.500M) | 23.3 | $--$ |
| | TD-error weighting | $0.715 \pm 0.090$ | $0.988 \pm 0.011$ | $0.800 \pm 0.447$ | 5/5 (0.610M) | 39.0 | $+0.038$ $[-0.015, 0.091]$ |
| DrawerClose–DDPG | SABER (Main) | $0.324 \pm 0.082$ | $0.277 \pm 0.100$ | $0.280 \pm 0.251$ | 5/5 (0.020M) | 24.0 | $--$ |
| | TD-error weighting | $0.313 \pm 0.098$ | $0.351 \pm 0.218$ | $0.400 \pm 0.166$ | 5/5 (0.130M) | 38.9 | $+0.011$ $[-0.079, 0.074]$ |
| FPP–SAC | SABER (Main) | $0.583 \pm 0.030$ | $0.986 \pm 0.011$ | $0.990 \pm 0.022$ | 5/5 (0.750M) | 29.1 | $--$ |
| | TD-error weighting | $0.544 \pm 0.068$ | $0.901 \pm 0.167$ | $0.990 \pm 0.022$ | 5/5 (0.800M) | 38.2 | $+0.040$ $[-0.016, 0.095]$ |

**Table 10:** Validated TD-error critic-loss weighting controls over paired seeds 1–5. Replay remains uniform, and only critic-loss coefficients change. Values are mean $\pm$ sample SD. The final column reports SABER-minus-control paired AUC differences with 95% bootstrap CIs; positive values favor SABER. For SAC, the priority signal is $\min(|\delta_{i1}|, |\delta_{i2}|)$, and the resulting normalized coefficient is applied to both critic losses. Steps/s is the descriptive seed mean defined in Section 1.3.

**Results.** SABER has higher observed mean AUC in all four validated cohorts, but every paired 95% interval includes zero. Late metrics are mixed, including higher FQ-AUC for TD-error weighting on DrawerClose.

**Interpretation.** The common mean direction distinguishes SABER from this specific TD-error allocation control, but the five-seed intervals do not establish a conclusive per-cohort advantage. TD3 is omitted because its stored control does not support the intended matched-signal comparison.

## 4.2 Action-Bin Resolution

**Purpose.** This sensitivity tests whether the Main action discretization $K = 5$ is robust to coarser and finer bins and records how bin resolution changes the frequency of support activation.

**Results.** Main $K = 5$ has higher mean AUC than both $K = 3$ and $K = 7$, with both Main-minus-variant 95% intervals excluding zero. The support-positive rate decreases as $K$ increases.

| Cohort | Method/Variant | AUC | FQ-AUC | Final | Hit90 | Steps/s | Support+ | ΔAUC |
|---|---|---|---|---|---|---|---|---|
| FPP–TD3 | $K = 3$ | $0.353 \pm 0.048$ | $0.603 \pm 0.121$ | $0.750 \pm 0.265$ | 2/5 (0.945M) | 33.9 | $0.396 \pm 0.012$ | $+0.089$ $[0.070, 0.109]$ |
| | $K = 5$ (Main) | $0.442 \pm 0.035$ | $0.804 \pm 0.103$ | $0.970 \pm 0.045$ | 5/5 (0.840M) | 41.2 | $0.219 \pm 0.009$ | $--$ |
| | $K = 7$ | $0.361 \pm 0.047$ | $0.616 \pm 0.151$ | $0.690 \pm 0.263$ | 2/5 (0.890M) | 36.1 | $0.148 \pm 0.010$ | $+0.081$ $[0.067, 0.093]$ |

**Table 11:** FPP TD3 action-bin sensitivity over paired seeds 1–5. Values are mean ± sample SD. Support+ is the support-positive rate. The final column reports Main-minus-variant paired AUC differences with 10,000-resample percentile-bootstrap 95% CIs; positive values favor Main $K = 5$.

**Interpretation.** The evaluated $K = 5$ setting is favored over both adjacent resolutions in this cohort. Because $K$ simultaneously changes action-bin geometry and activation density, the experiment does not identify either factor alone or establish $K = 5$ as universally optimal.

### 4.3 Actor and Critic Weight Scope

**Purpose.** This ablation tests whether the support coefficient should modify the critic, the actor, or both. All variants retain the support query, support hyperparameters, and minibatch normalization. Critic-Only weights Bellman losses; Actor-Only weights per-sample actor terms; Actor+Critic weights both.

| Cohort | Method/Variant | AUC | FQ-AUC | Final | Hit90 | Steps/s | ΔAUC (Main–variant) |
|---|---|---|---|---|---|---|---|
| FPP–DDPG | Critic-Only (Main) | $0.609 \pm 0.038$ | $0.961 \pm 0.014$ | $0.960 \pm 0.042$ | 5/5 (0.540M) | 41.7 | $--$ |
| | Actor-Only | $0.589 \pm 0.032$ | $0.952 \pm 0.016$ | $0.960 \pm 0.022$ | 5/5 (0.600M) | 15.8 | $+0.020[-0.024, 0.063]$ |
| | Actor+Critic | | | *pending* | | | |
| DrawerOpen–DDPG | Critic-Only (Main) | $0.754 \pm 0.044$ | $0.987 \pm 0.018$ | $1.000 \pm 0.000$ | 5/5 (0.500M) | 23.3 | $--$ |
| | Actor-Only | $0.612 \pm 0.343$ | $0.760 \pm 0.433$ | $0.800 \pm 0.447$ | 4/5 (0.470M) | 18.6 | $+0.141[-0.051, 0.478]$ |
| | Actor+Critic | $0.728 \pm 0.084$ | $0.992 \pm 0.008$ | $0.990 \pm 0.022$ | 5/5 (0.540M) | 18.4 | $+0.026[-0.048, 0.091]$ |
| DrawerClose–DDPG | Critic-Only (Main) | $0.324 \pm 0.082$ | $0.277 \pm 0.100$ | $0.280 \pm 0.251$ | 5/5 (0.020M) | 24.0 | $--$ |
| | Actor-Only | $0.307 \pm 0.039$ | $0.299 \pm 0.061$ | $0.280 \pm 0.299$ | 5/5 (0.040M) | 24.0 | $+0.017[-0.026, 0.078]$ |
| | Actor+Critic | $0.298 \pm 0.071$ | $0.241 \pm 0.119$ | $0.440 \pm 0.370$ | 5/5 (0.050M) | 23.7 | $+0.026[-0.043, 0.091]$ |

**Table 12:** DDPG actor-scope ablations over paired seeds 1–5. Values are mean ± sample standard deviation. Hit90 reports the number of reaching seeds and the reached-seed median step. Steps/s is the descriptive seed mean defined in Section 1.3. The final column gives Main-minus-variant paired AUC differences with 95% bootstrap confidence intervals. Pending cells are left unreported until a complete five-seed cohort is available.

**Results.** Critic-Only has the highest observed mean AUC in each of the five completed Main-versus-scope comparisons. Main-minus-variant differences range from $+0.017$ to $+0.141$, but every 95% interval includes zero. DrawerOpen Actor-Only also has substantial seed dispersion $(0.612 \pm 0.343)$. FPP Actor+Critic remains pending and is not inferred.

**Interpretation.** The available results are consistent with using critic-only weighting as the conservative main design, while providing no decisive evidence that actor weighting is intrinsically harmful. Each scope changes the subsequent training trajectory, so realized support masks need not remain identical.

### 4.4 Secondary Replay and Update-Budget References

**Purpose.** These references place SABER outside the strictly matched UTD-1 comparison: no-HER replay variants test whether replay filtering alone solves FPP, while high-replay-ratio HER and REDQ-HER test larger update budgets and critic ensembles. REDQ-HER TQB clips the sampled target-critic backup to the sparse-reward range $[-50, 0]$ before forming the Bellman target.

**Results.** Without HER, the best mean AUC is $0.148$ on FPP DDPG, $0.039$ on FPP TD3, and $0.064$ on FPP SAC. On FPP SAC, high-replay-ratio HER and REDQ-HER reach AUC $0.768$ and $0.867$, respectively, above SABER's $0.583$, but use UTD 4 and REDQ also uses ten critics. On DrawerClose, REDQ-HER reaches Hit90 in all seeds at 20k but records AUC/FQ-AUC $0.188/0.098$, compared with $0.324/0.277$ for DDPG SABER.

**Interpretation.** The no-HER runs show that replay filtering in the evaluated forms does not replace hindsight relabeling on these sparse-reward tasks. Increased update budget and ensemble size can be stronger on FPP SAC but are not uniformly stronger across tasks. Because HER use, UTD, algorithm, and architecture differ, these tables are scaling and context references rather than one-factor tests of SABER.

| Cohort | Method/Variant | AUC | FQ-AUC | Final | Hit90 | Steps/s |
|---|---|---|---|---|---|---|
| FPP–DDPG | ER | $0.148 \pm 0.039$ | $0.354 \pm 0.124$ | $0.460 \pm 0.188$ | 0/5 | 68.4 |
| | Annealing ER | $0.129 \pm 0.111$ | $0.259 \pm 0.251$ | $0.340 \pm 0.363$ | 1/5 (0.960M) | 68.0 |
| | AER | $0.044 \pm 0.002$ | $0.045 \pm 0.004$ | $0.000 \pm 0.000$ | 0/5 | 67.0 |
| FPP–TD3 | ER | $0.039 \pm 0.002$ | $0.043 \pm 0.002$ | $0.000 \pm 0.000$ | 0/5 | 21.0 |
| | Annealing ER | $0.039 \pm 0.002$ | $0.043 \pm 0.002$ | $0.000 \pm 0.000$ | 0/5 | 19.6 |
| | AER | $0.039 \pm 0.002$ | $0.043 \pm 0.002$ | $0.000 \pm 0.000$ | 0/5 | 26.2 |
| FPP–SAC | ER | $0.064 \pm 0.042$ | $0.114 \pm 0.135$ | $0.130 \pm 0.199$ | 0/5 | 39.4 |
| | Annealing ER | $0.041 \pm 0.002$ | $0.047 \pm 0.008$ | $0.000 \pm 0.000$ | 0/5 | 39.5 |
| | AER | $0.040 \pm 0.003$ | $0.044 \pm 0.002$ | $0.000 \pm 0.000$ | 0/5 | 39.3 |

**Table 13:** FPP no-HER replay references over seeds 1–5. Values are mean ± sample SD on the exact 10k–1M grid. These runs omit hindsight relabeling and are context references rather than matched post-HER mechanism controls. Steps/s is the descriptive seed mean defined in Section 1.3.

| Cohort | Method/Variant | AUC | FQ-AUC | Final | Hit90 | Steps/s |
|---|---|---|---|---|---|---|
| FPP–SAC | HER<br>UTD 1; 2 critics | $0.471 \pm 0.079$ | $0.735 \pm 0.247$ | $0.790 \pm 0.288$ | 3/5 (0.800M) | 38.9 |
| | SABER<br>UTD 1; 2 critics | $0.583 \pm 0.030$ | $0.986 \pm 0.011$ | $0.990 \pm 0.022$ | 5/5 (0.750M) | 29.1 |
| | HER high-RR<br>UTD 4; 2 critics | $0.768 \pm 0.097$ | $0.963 \pm 0.021$ | $0.960 \pm 0.042$ | 5/5 (0.450M) | 5.9 |
| | REDQ-HER TQB | | | | | |
| FPP–REDQ | UTD 4; 10 critics/2 target | $0.867 \pm 0.052$ | $0.993 \pm 0.008$ | $1.000 \pm 0.000$ | 5/5 (0.300M) | 3.1 |

**Table 14:** FPP SAC and REDQ scaling references on the exact shared 50k–1M grid. Values are mean ± sample SD over seeds 1–5. High-replay-ratio HER and REDQ-HER change the update budget, and REDQ-HER also changes critic architecture; they are not matched post-HER loss-allocation controls. Steps/s is the descriptive seed mean defined in Section 1.3.

| Cohort | Method/Variant | AUC | FQ-AUC | Final | Hit90 | Steps/s |
|---|---|---|---|---|---|---|
| DrawerClose–DDPG | HER<br>UTD 1; 1 critic | $0.251 \pm 0.030$ | $0.250 \pm 0.093$ | $0.160 \pm 0.182$ | 5/5 (0.070M) | 41.2 |
| | SABER<br>UTD 1; 1 critic | $0.324 \pm 0.082$ | $0.277 \pm 0.100$ | $0.280 \pm 0.251$ | 5/5 (0.020M) | 24.0 |
| | REDQ-HER TQB | | | | | |
| DrawerClose–REDQ | UTD 4; 10 critics/2 target | $0.188 \pm 0.041$ | $0.098 \pm 0.022$ | $0.100 \pm 0.117$ | 5/5 (0.020M) | 5.8 |

**Table 15:** DrawerClose update-budget and architecture reference over seeds 1–5. Values are mean ± sample standard deviation on the exact 10k–2M grid. FQ-AUC covers 1.5M–2M; Final is final-checkpoint success; Hit90 gives reaching seeds and their median first-hit step. REDQ-HER uses four updates per environment step and a 10-critic ensemble with two target critics sampled per backup, so it is not a resource-matched SABER comparison. Steps/s is the descriptive seed mean defined in Section 1.3.

# 5 Support-Signal Diagnostic Details

## 5.1 Instrumentation and Aggregation

**Purpose.** The diagnostics test whether SABER's support rule activates in practice, whether it is redundant with TD-error ranking, and whether the resulting weights alter immediate critic fitting in the intended direction.

**Protocol.** Diagnostic logging introduces no gradient term and does not change replay sampling, objectives, or parameter-update order. FPP logs one full minibatch every 10k environment steps for DDPG, TD3, and SAC seeds 1–5: 100 batches per run, 500 per backbone, and 1,500 total. Drawer diagnostics use a 200-step cadence from 5k through 2M, producing 9,976 persisted snapshots per seed. Snapshot quantities are aggregated within seed before cohort statistics are computed.

For each persisted minibatch, the diagnostic stores the pre-update absolute Bellman residual, support membership, and the residual after one critic update. The fixed-target residual is

$$\Delta|\delta_i| = \left|Q_\phi^{\mathrm{pre}}(x_i, a_i^r) - y_i^{\mathrm{pre}}\right| - \left|Q_\phi^{\mathrm{post}}(x_i, a_i^r) - y_i^{\mathrm{pre}}\right|.$$

Thus the target and minibatch are held fixed while the critic parameters change. Live-target quantities, where available, are retained only for audit and are not mixed with the fixed-target statistic.

**Interpretation boundary.** These measurements audit activation and one-update optimization behavior on persisted minibatches. They do not by themselves identify the independent cause of a long-horizon return difference.

## 5.2 Cohort-Level Results

**Purpose.** The cohort summary jointly measures activation frequency, association with pre-update error, and support-conditioned residual reduction under a common fixed-target definition.

**Results.** Primary-cohort support-positive rates range from $0.197$ to $0.417$, so the indicator is neither nearly absent nor nearly constant. Primary Pearson correlations are modest ($0.084$–$0.143$), while exact top-decile enrichment ranges from $1.379\times$ to $1.828\times$.

| Cohort | Seeds | Support+ rate | Pearson $r$ | Top-10% enrich. | Pre-update $\lvert\delta\rvert$ $(+/-)$ | One-step $\Delta\lvert\delta\rvert$ $(+/-)$ |
|---|---|---|---|---|---|---|
| FPP–DDPG | 1–5 | $0.367 \pm 0.011$ | 0.116 | $1.409\times$ | $1.42\times$ | $2.39\times$ |
| FPP–TD3 | 1–5 | $0.219 \pm 0.009$ | 0.128 | $1.670\times$ | $1.62\times$ | $3.30\times$ |
| FPP–SAC | 1–5 | $0.197 \pm 0.013$ | 0.143 | $1.828\times$ | $1.74\times$ | $3.94\times$ |
| DrawerOpen–DDPG | 1–5 | $0.417 \pm 0.021$ | 0.084 | $1.379\times$ | $1.16\times$ | $1.79\times$ |
| DrawerOpen–DDPG | 6–10 extension | $0.475 \pm 0.094$ | 0.095 | $1.440\times$ | $1.13\times$ | $1.91\times$ |
| DrawerOpen–DDPG | 1–10 pooled | $0.446 \pm 0.071$ | 0.090 | $1.409\times$ | $1.15\times$ | $1.83\times$ |
| DrawerClose–DDPG | 1–5 | $0.298 \pm 0.018$ | 0.104 | $1.534\times$ | $1.42\times$ | $2.94\times$ |
| DrawerClose–DDPG | 6–10 extension | $0.299 \pm 0.007$ | 0.118 | $1.602\times$ | $1.48\times$ | $4.23\times$ |
| DrawerClose–DDPG | 1–10 pooled | $0.298 \pm 0.013$ | 0.111 | $1.568\times$ | $1.45\times$ | $3.48\times$ |

**Table 16:** Support-signal diagnostics. Statistics are first aggregated within seed, after which seeds are independent units. Support rates are mean $\pm$ sample SD. The last two columns divide support-positive by support-negative cohort means. One-step $\Delta|\delta|$ evaluates the same minibatch after one critic step while retaining the pre-update Bellman target. Seeds 1–5 are the primary diagnostic cohorts, seeds 6–10 are separate extensions, and pooled ten-seed rows are descriptive.

The support-positive/support-negative one-step residual-reduction ratio ranges from $1.79\times$ to $3.94\times$ in the primary cohorts. Both Drawer extensions preserve enrichment above one and a larger reduction for support-positive samples.

**Interpretation.** Support-positive samples are enriched among high-error samples, but the weak linear correlations show that support does not reduce to a TD-error ranking. The one-step statistic verifies that the normalized coefficients redistribute immediate critic fitting as intended; it is not a standalone explanation of long-term learning performance.

### 5.3 Phase-Conditioned Allocation

**Purpose.** The phase analysis asks where support activation occurs during Drawer progression and whether it merely follows the most frequent phase.

| Cohort | Phase | Sample fraction | Support+ rate | Seeds |
|---|---|---|---|---|
| DrawerOpen–DDPG | Approach | $0.174 \pm 0.018$ | $0.745 \pm 0.022$ | 1–5 |
| | Contact | $0.049 \pm 0.003$ | $0.698 \pm 0.056$ | |
| | Opening | $0.046 \pm 0.005$ | $0.636 \pm 0.053$ | |
| | Open | $0.730 \pm 0.018$ | $0.306 \pm 0.027$ | |
| DrawerOpen–DDPG | Approach | $0.195 \pm 0.011$ | $0.754 \pm 0.030$ | 6–10 ext. |
| | Contact | $0.027 \pm 0.024$ | $0.515 \pm 0.181$ | |
| | Opening | $0.023 \pm 0.023$ | $0.440 \pm 0.263$ | |
| | Open | $0.755 \pm 0.039$ | $0.387 \pm 0.127$ | |
| DrawerClose–DDPG | Approach | $0.218 \pm 0.002$ | $0.549 \pm 0.032$ | 1–5 |
| | Contact | $0.016 \pm 0.003$ | $0.710 \pm 0.026$ | |
| | Closing | $0.019 \pm 0.004$ | $0.425 \pm 0.034$ | |
| | Closed | $0.747 \pm 0.004$ | $0.212 \pm 0.017$ | |
| DrawerClose–DDPG | Approach | $0.222 \pm 0.003$ | $0.569 \pm 0.004$ | 6–10 ext. |
| | Contact | $0.018 \pm 0.003$ | $0.727 \pm 0.016$ | |
| | Closing | $0.017 \pm 0.002$ | $0.438 \pm 0.028$ | |
| | Closed | $0.743 \pm 0.007$ | $0.204 \pm 0.010$ | |

**Table 17:** Drawer phase-conditioned allocation. Phases are defined from the relabeled training goal. Values are seed means $\pm$ sample SD; support is sample-weighted within each seed. Seeds 6–10 are separate extensions. Pooled ten-seed and seed-level values are included in the artifacts.

**Results.** In both Drawer tasks and in both primary and extension cohorts, approach and contact have higher support-positive rates than the dominant terminal phase, even though the terminal phase accounts for roughly three quarters of logged samples.

**Interpretation.** Activation therefore does not simply track phase frequency. The task-specific phase labels localize the signal but do not constitute an independent causal test of long-horizon performance.

## 6 Runtime and Resource Measurements

### 6.1 Execution-Environment Audit

**Purpose.** The audit records the execution boundary needed to assess paired fairness and to distinguish software-controlled comparisons from hardware-specific throughput observations.

**Results.** All four hosts used the verified common experiment-library stack: Python 3.12.3, PyTorch 2.12.1+cu130, CUDA 13.0, cuDNN 9.20.0, Gymnasium 1.3.0, Gymnasium-Robotics 1.4.2, MuJoCo 3.3.0, Meta-World 3.1.1, NumPy 2.3.5, and SciPy 1.18.0. For each paired HER–SABER cohort, both methods and all paired seeds were executed on one physical host under this stack; no paired cohort was split across machines. Sanitized provenance is recorded in `execution_host_mapping.csv`.

| Host | System | Compute hardware | OS, kernel, and driver |
|---|---|---|---|
| A | MSI GB10 system | 20-core ARM (10 Cortex-X925 + 10 Cortex-A725), 121 GiB unified memory, NVIDIA GB10 | Ubuntu 24.04.4; 6.17.0-1014-nvidia; driver 580.142 |
| B | GIGABYTE AI TOP ATOM | 20-core ARM (10 Cortex-X925 + 10 Cortex-A725), 121 GiB unified memory, NVIDIA GB10 | Ubuntu 24.04.4; 6.17.0-1026-nvidia; driver 580.159.03 |
| C | GIGABYTE AI TOP ATOM | 20-core ARM (10 Cortex-X925 + 10 Cortex-A725), 121 GiB unified memory, NVIDIA GB10 | Ubuntu 24.04.4; 6.17.0-1026-nvidia; driver 580.159.03 |
| D | GB10 system | 20-core ARM (10 Cortex-X925 + 10 Cortex-A725), 121 GiB unified memory, NVIDIA GB10 | Ubuntu 24.04.3; 6.11.0-1016-nvidia; driver 580.95.05 |

**Table 18:** Hardware inventory for the four physically distinct execution hosts. All systems use CUDA-addressable unified memory rather than dedicated VRAM. Hosts B and C are separate machines with the same model and system configuration. Hostnames, serial numbers, and other identifying metadata are omitted.

**Interpretation.** The paired performance differences are not created by moving HER and SABER across machines within a cohort. Hardware still differs across cohorts, so historical throughput is descriptive and is not used for cross-host ranking.

### 6.2 Dedicated Runtime Profile

**Purpose.** The dedicated profile quantifies the current implementation overhead under a sequential same-hardware comparison and identifies which pipeline component dominates that overhead. It runs HER and SABER for FPP TD3 seed 1 over the 20k–100k interval.

| Profile item | Recorded value |
|---|---|
| Execution protocol | Sequential FPP TD3 HER and SABER, seed 1, 20k–100k |
| GPU | NVIDIA GB10, compute capability 12.1, 121.7 GiB CUDA-addressable memory |
| CPU/platform | 20 logical CPUs, AArch64 Linux 6.17.0-1026-nvidia, glibc 2.39 |
| Software | Python 3.12.3, PyTorch 2.12.1+cu130, CUDA 13.0, cuDNN 9.20.0 |
| Comparison boundary | The dedicated profile is not extrapolated to historical primary runs |
| Memory boundary | Peak process and accelerator memory were not measured |

**Table 19:** Recorded infrastructure for the dedicated same-hardware runtime profile. CUDA-addressable memory is reported because the GB10 uses unified memory. These identifiers characterize the dedicated runtime profile; historical steps/s values in other tables are descriptive cohort measurements.

| Component | HER ms/step | SABER ms/step | Difference | SABER share |
|---|---|---|---|---|
| Interaction | 1.000 | 1.071 | $+0.070$ | 4.4% |
| Replay + HER | 1.925 | 2.651 | $+0.726$ | 11.0% |
| Batch preparation | 0.096 | 0.108 | $+0.012$ | 0.4% |
| Support and weighting | 0.000 | 15.028 | $+15.028$ | 62.3% |
| Critic update | 2.945 | 2.832 | $-0.113$ | 11.7% |
| Actor + target update | 1.384 | 1.292 | $-0.091$ | 5.4% |
| Evaluation/logging amortized | 0.080 | 0.085 | $+0.005$ | 0.4% |
| Other loop overhead | 0.633 | 1.066 | $+0.434$ | 4.4% |
| Total training loop | 8.063 | 24.133 | $+16.070$ | 100.0% |

**Table 20:** Hardware-specific component times for the FPP TD3 seed-1 implementation profile over 20k–100k steps. Other loop overhead is the difference between total loop time and the explicitly instrumented components.

**Results.** HER and SABER achieve 124.0 and 41.4 environment steps/s, corresponding to 8.063 and 24.133 ms per training step. Support and weighting account for 15.028 ms/step, or 62.3% of SABER's measured loop time, and explain most of the $+16.070$ ms/step difference.

**Interpretation.** The profile attributes the dominant measured cost to the current support retrieval implementation, motivating faster retrieval. It is one hardware-specific, early-training profile rather than an asymptotic or end-to-end learning-efficiency result. Historical steps/s values were not collected under this sequential protocol, and peak memory was not measured; neither quantity is inferred.

## 7 Artifact and Provenance Index

**Purpose.** This index connects each reported experiment to a validation status and repository-relative artifact location, so completed evidence is distinguishable from references, extensions, and pending work.

| Content | Artifact key | Status |
|---|---|---|
| Primary seed metrics and curves | `primary_seed_results` | Complete |
| Action-bin sensitivity | `action_bin_sensitivity` | Complete |
| DDPG actor-scope analysis | `actor_scope_ddpg` | FPP A+C pending |
| Derived Technical Supplement summaries | `technical_supplement` | FPP A+C pending |
| FPP diagnostics | `fpp_diagnostic` | Complete |
| DrawerClose diagnostics | `drawerclose_diagnostic` | Complete |
| DrawerOpen fixed-target diagnostics | `draweropen_fixed_target` | Complete |
| FPP TD3 runtime profile | `fpp_td3_runtime` | Complete |
| Execution-environment audit | `execution_environment_audit` | Complete |
| FPP TD3 ten-seed confirmation | `fpp_td3_n10` | Complete |
| FPP TD3 inclusive-future sensitivity | `fpp_td3_inclusive` | Complete |
| FPP SAC update-budget references | `fpp_sac_scaling` | Complete |
| DrawerClose REDQ scaling reference | `drawerclose_redq` | Complete |

**Table 21:** Artifact index. Keys resolve to repository-relative package paths in `artifact_index.csv`. Canonical artifacts include manifests that bind source files by repository-relative path and SHA256.

**Development scope and seed mapping.** We did not conduct an exhaustive joint hyperparameter search. The Main SABER settings were fixed for the matched comparisons, and no control received variant-specific tuning. The $K \in \{3, 7\}$ runs are sensitivity checks around Main $K = 5$, not a per-task model-selection sweep. For paired seed $s$, the environment, PyTorch, NumPy, and action-noise streams use $s$; the support-weight stream uses $20000 + s$, and evaluation uses $10000 + s$. Training and evaluation resets advance by deterministic episode- and checkpoint-indexed offsets. All methods in a paired cohort use this mapping.

**Interpretation boundary.** Generated analysis directories contain compact CSV summaries, a repository-relative source manifest, and a self-excluding artifact manifest. Canonical experiment artifacts retain configuration, environment snapshot, source provenance, and validation records where available. Raw result files are referenced rather than duplicated in the derived summaries. An indexed artifact establishes traceability to the reported result; it does not convert a secondary reference into a matched causal comparison.

# A Primary Seed-Level Results

**Purpose.** The seed-level appendix exposes every observation underlying the primary aggregate table, allowing readers to inspect paired directions and between-seed dispersion directly.

**Results.** SABER has higher observed mean AUC than matched HER in all seven primary cohorts, and four paired 95% intervals exclude zero. The individual rows show that this aggregate statement coexists with negative or near-zero seed pairs and especially large dispersion on the Drawer TD3 cohorts. AUC and final-quarter AUC use each cohort's exact common grid.

**Interpretation.** These rows are the observations used in the primary analysis, not an additional independent experiment. They make the direction and uncertainty of each paired cohort auditable without changing the seed as the unit of statistical analysis.

| Method | Seed | AUC | FQ-AUC | Final | Hit90 step |
|---|---|---|---|---|---|
| HER | 1 | 0.640152 | 0.956000 | 0.900 | 0.510M |
| | 2 | 0.537374 | 0.934000 | 0.950 | 0.600M |
| | 3 | 0.578788 | 0.964000 | 0.950 | 0.610M |
| | 4 | 0.325253 | 0.731000 | 0.900 | 0.920M |
| | 5 | 0.612374 | 0.967000 | 1.000 | 0.570M |
| SABER | 1 | 0.656061 | 0.939000 | 0.950 | 0.480M |
| | 2 | 0.615152 | 0.972000 | 0.950 | 0.540M |
| | 3 | 0.564899 | 0.959000 | 1.000 | 0.630M |
| | 4 | 0.631818 | 0.976000 | 1.000 | 0.480M |
| | 5 | 0.577525 | 0.960000 | 0.900 | 0.610M |
| HGR | 1 | 0.441667 | 0.765000 | 1.000 | 0.870M |
| | 2 | 0.076515 | 0.065000 | 0.000 | – |
| | 3 | 0.374495 | 0.482000 | 0.500 | – |
| | 4 | 0.417677 | 0.587000 | 0.700 | 0.990M |
| | 5 | 0.380556 | 0.506000 | 0.500 | – |
| PER | 1 | 0.484091 | 0.919000 | 1.000 | 0.770M |
| | 2 | 0.520455 | 0.953000 | 1.000 | 0.730M |
| | 3 | 0.604293 | 0.946000 | 0.900 | 0.550M |
| | 4 | 0.643434 | 0.964000 | 1.000 | 0.460M |
| | 5 | 0.049242 | 0.046000 | 0.000 | – |

**Table 22:** Seed-level primary results for FPP with DDPG. Metrics use the exact shared checkpoint grid without smoothing or interpolation.

| Method | Seed | AUC | FQ-AUC | Final | Hit90 step |
|---|---|---|---|---|---|
| HER | 1 | 0.346465 | 0.523000 | 0.550 | – |
| | 2 | 0.372727 | 0.635000 | 0.950 | 1.000M |
| | 3 | 0.394192 | 0.662000 | 1.000 | 0.980M |
| | 4 | 0.410859 | 0.686000 | 0.950 | 0.970M |
| | 5 | 0.333838 | 0.530000 | 0.750 | – |
| SABER | 1 | 0.430051 | 0.846000 | 1.000 | 0.840M |
| | 2 | 0.456818 | 0.856000 | 1.000 | 0.860M |
| | 3 | 0.495960 | 0.914000 | 0.950 | 0.630M |
| | 4 | 0.406566 | 0.652000 | 0.900 | 0.630M |
| | 5 | 0.422475 | 0.752000 | 1.000 | 0.910M |
| HGR | 1 | 0.345202 | 0.515000 | 0.500 | – |
| | 2 | 0.396212 | 0.593000 | 0.800 | – |
| | 3 | 0.407576 | 0.581000 | 0.850 | – |
| | 4 | 0.326515 | 0.489000 | 0.450 | 0.630M |
| | 5 | 0.361111 | 0.493000 | 0.450 | – |
| PER | 1 | 0.345455 | 0.515000 | 0.550 | – |
| | 2 | 0.408586 | 0.653000 | 0.950 | 1.000M |
| | 3 | 0.394192 | 0.628000 | 0.800 | – |
| | 4 | 0.369192 | 0.526000 | 0.500 | – |
| | 5 | 0.327020 | 0.488000 | 0.500 | – |

**Table 23:** Seed-level primary results for FPP with TD3. Metrics use the exact shared checkpoint grid without smoothing or interpolation.

| Method | Seed | AUC | FQ-AUC | Final | Hit90 step |
|---|---|---|---|---|---|
| HER | 1 | 0.559211 | 0.985000 | 1.000 | 0.800M |
| | 2 | 0.469737 | 0.760000 | 1.000 | 0.950M |
| | 3 | 0.406579 | 0.500000 | 0.500 | – |
| | 4 | 0.540789 | 0.965000 | 1.000 | 0.800M |
| | 5 | 0.378947 | 0.465000 | 0.450 | – |
| SABER | 1 | 0.636842 | 0.995000 | 0.950 | 0.650M |
| | 2 | 0.567105 | 0.980000 | 1.000 | 0.800M |
| | 3 | 0.573684 | 0.980000 | 1.000 | 0.750M |
| | 4 | 0.575000 | 1.000000 | 1.000 | 0.750M |
| | 5 | 0.564474 | 0.975000 | 1.000 | 0.800M |
| HGR | 1 | 0.451316 | 0.545000 | 0.800 | – |
| | 2 | 0.596053 | 0.980000 | 1.000 | 0.700M |
| | 3 | 0.506579 | 0.800000 | 1.000 | 0.900M |
| | 4 | 0.450000 | 0.505000 | 0.500 | – |
| | 5 | 0.414474 | 0.490000 | 0.500 | – |
| PER | 1 | 0.755263 | 0.970000 | 1.000 | 0.450M |
| | 2 | 0.635526 | 0.985000 | 1.000 | 0.550M |
| | 3 | 0.706579 | 0.990000 | 1.000 | 0.550M |
| | 4 | 0.690789 | 1.000000 | 1.000 | 0.550M |
| | 5 | 0.660526 | 0.985000 | 0.950 | 0.600M |

**Table 24:** Seed-level primary results for FPP with SAC. Metrics use the exact shared checkpoint grid without smoothing or interpolation.

| Method | Seed | AUC | FQ-AUC | Final | Hit90 step |
|---|---|---|---|---|---|
| HER | 1 | 0.460302 | 0.970000 | 1.000 | 1.090M |
| | 2 | 0.000000 | 0.000000 | 0.000 | – |
| | 3 | 0.000000 | 0.000000 | 0.000 | – |
| | 4 | 0.000000 | 0.000000 | 0.000 | – |
| | 5 | 0.576256 | 0.993500 | 0.950 | 0.830M |
| SABER | 1 | 0.757286 | 1.000000 | 1.000 | 0.510M |
| | 2 | 0.786935 | 0.977000 | 1.000 | 0.420M |
| | 3 | 0.705025 | 1.000000 | 1.000 | 0.580M |
| | 4 | 0.713065 | 0.960000 | 1.000 | 0.500M |
| | 5 | 0.806030 | 1.000000 | 1.000 | 0.400M |
| HGR | 1 | 0.799749 | 0.995000 | 0.500 | 0.410M |
| | 2 | 0.704523 | 0.990000 | 1.000 | 0.530M |
| | 3 | 0.692211 | 0.993000 | 1.000 | 0.610M |
| | 4 | 0.794975 | 0.998000 | 1.000 | 0.430M |
| | 5 | 0.449497 | 0.934500 | 1.000 | 1.060M |
| PER | 1 | 0.859799 | 1.000000 | 1.000 | 0.330M |
| | 2 | 0.877387 | 0.998000 | 1.000 | 0.270M |
| | 3 | 0.814070 | 0.969000 | 1.000 | 0.330M |
| | 4 | 0.809171 | 0.951500 | 0.950 | 0.390M |
| | 5 | 0.765075 | 0.990000 | 1.000 | 0.410M |

**Table 25:** Seed-level primary results for DrawerOpen with DDPG. Metrics use the exact shared checkpoint grid without smoothing or interpolation.

| Method | Seed | AUC | FQ-AUC | Final | Hit90 step |
|---|---|---|---|---|---|
| HER | 1 | 0.632286 | 0.999500 | 0.950 | 0.740M |
| | 2 | 0.000000 | 0.000000 | 0.000 | – |
| | 3 | 0.000000 | 0.000000 | 0.000 | – |
| | 4 | 0.000251 | 0.000000 | 0.000 | – |
| | 5 | 0.000000 | 0.000000 | 0.000 | – |
| SABER | 1 | 0.486558 | 0.996500 | 0.950 | 1.070M |
| | 2 | 0.000000 | 0.000000 | 0.000 | – |
| | 3 | 0.000503 | 0.000000 | 0.000 | – |
| | 4 | 0.689950 | 0.997000 | 1.000 | 0.560M |
| | 5 | 0.632161 | 0.997000 | 1.000 | 0.780M |
| HGR | 1 | 0.000000 | 0.000000 | 0.000 | – |
| | 2 | 0.692462 | 0.992000 | 1.000 | 0.520M |
| | 3 | 0.531910 | 0.997000 | 0.900 | 0.930M |
| | 4 | 0.000000 | 0.000000 | 0.000 | – |
| | 5 | 0.628266 | 0.996000 | 1.000 | 0.760M |
| PER | 1 | 0.632412 | 0.986000 | 1.000 | 0.780M |
| | 2 | 0.000000 | 0.000000 | 0.000 | – |
| | 3 | 0.000000 | 0.000000 | 0.000 | – |
| | 4 | 0.755528 | 0.999000 | 1.000 | 0.520M |
| | 5 | 0.644221 | 0.994000 | 1.000 | 0.720M |

**Table 26:** Seed-level primary results for DrawerOpen with TD3. Metrics use the exact shared checkpoint grid without smoothing or interpolation.

| Method | Seed | AUC | FQ-AUC | Final | Hit90 step |
|---|---|---|---|---|---|
| HER | 1 | 0.281658 | 0.337500 | 0.300 | 0.090M |
| | 2 | 0.268844 | 0.315500 | 0.100 | 0.150M |
| | 3 | 0.203518 | 0.193000 | 0.400 | 0.040M |
| | 4 | 0.249497 | 0.289000 | 0.000 | 0.070M |
| | 5 | 0.251508 | 0.116000 | 0.000 | 0.020M |
| SABER | 1 | 0.287814 | 0.165500 | 0.650 | 0.010M |
| | 2 | 0.337437 | 0.417000 | 0.200 | 0.060M |
| | 3 | 0.284045 | 0.306500 | 0.150 | 0.050M |
| | 4 | 0.461307 | 0.300500 | 0.000 | 0.020M |
| | 5 | 0.251508 | 0.197500 | 0.400 | 0.010M |
| HGR | 1 | 0.672613 | 0.218000 | 0.000 | 0.100M |
| | 2 | 0.323869 | 0.236500 | 0.000 | 0.090M |
| | 3 | 0.237563 | 0.154500 | 0.150 | 0.120M |
| | 4 | 0.238568 | 0.219500 | 0.050 | 0.100M |
| | 5 | 0.273367 | 0.299000 | 0.200 | 0.010M |
| PER | 1 | 0.288568 | 0.272000 | 0.550 | 0.100M |
| | 2 | 0.244975 | 0.273500 | 0.000 | 0.180M |
| | 3 | 0.297990 | 0.392000 | 1.000 | 0.040M |
| | 4 | 0.331281 | 0.356500 | 0.350 | 0.050M |
| | 5 | 0.376884 | 0.404000 | 0.400 | 0.060M |

**Table 27:** Seed-level primary results for DrawerClose with DDPG. Metrics use the exact shared checkpoint grid without smoothing or interpolation.

| Method | Seed | AUC | FQ-AUC | Final | Hit90 step |
|---|---|---|---|---|---|
| HER | 1 | 0.450628 | 0.455000 | 0.850 | 0.080M |
| | 2 | 0.058291 | 0.002000 | 0.000 | 0.120M |
| | 3 | 0.886181 | 0.847000 | 0.900 | 0.050M |
| | 4 | 0.097236 | 0.017000 | 0.000 | 0.060M |
| | 5 | 0.151382 | 0.098000 | 0.000 | 0.070M |
| SABER | 1 | 0.080653 | 0.020000 | 0.000 | 0.080M |
| | 2 | 0.908291 | 0.978000 | 1.000 | 0.090M |
| | 3 | 0.739950 | 0.761000 | 1.000 | 0.200M |
| | 4 | 0.908920 | 0.909500 | 0.950 | 0.010M |
| | 5 | 0.106030 | 0.000000 | 0.000 | 0.050M |
| HGR | 1 | 0.923618 | 0.886000 | 0.400 | 0.080M |
| | 2 | 0.686307 | 0.808500 | 0.950 | 0.060M |
| | 3 | 0.782161 | 0.428000 | 1.000 | 0.130M |
| | 4 | 0.477010 | 0.115000 | 0.150 | 0.060M |
| | 5 | 0.138945 | 0.063000 | 0.000 | 0.260M |
| PER | 1 | 0.111055 | 0.050000 | 0.000 | 0.220M |
| | 2 | 0.845477 | 0.987000 | 1.000 | 0.190M |
| | 3 | 0.824874 | 0.891000 | 1.000 | 0.120M |
| | 4 | 0.112060 | 0.002000 | 0.000 | 0.120M |
| | 5 | 0.725377 | 0.983000 | 1.000 | 0.100M |

**Table 28:** Seed-level primary results for DrawerClose with TD3. Metrics use the exact shared checkpoint grid without smoothing or interpolation.